

The `getType()` method ❷ just returns the type of the shape given at initialization.

We can now write any number of subclasses that inherit from `Shape`. We'll create three subclasses that will call the `__init__()` method of the `Shape` class, passing in a string that identifies its type and the size of the window. The `getType()` method will only appear in the `Shape` class, so any client calls to `getType()` will be handled by that method in the inherited `Shape` class. We'll start with the code for the `Square` class, shown in [Listing 10-7](#).

File: InheritedShapes/Square.py

```
# Square class

import pygame
from Shape import *

class Square(Shape): ❶

    def __init__(self, window, maxWidth, maxHeight):
        super().__init__(window, 'Square', maxWidth, maxHeight) ❷
        self.widthAndHeight = random.randrange(10, 100)
        self.rect = pygame.Rect(self.x, self.y,
                                self.widthAndHeight, self.widthAndHeight)

    def clickedInside(self, mousePoint): ❸
        clicked = self.rect.collidepoint(mousePoint)
        return clicked

    def getArea(self): ❹
        theArea = self.widthAndHeight * self.widthAndHeight
        return theArea

    def draw(self): ❺
        pygame.draw.rect(self.window, self.color,
                         (self.x, self.y, self.widthAndHeight,
                          self.widthAndHeight))
```

Listing 10-7: The `Square` class that inherits from the `Shape` class